

Test of Mathematics for University Admission, 2021 Paper 1

Worked Solutions

Version 1.0, March 2022

Contents

Introduction for students	2
Question 1	3
Question 2	4
Question 3	5
Question 4	6
Question 5	7
Question 6	8
Question 7	9
Question 8	10
Question 9	11
Question 10	12
Question 11	13
Question 12	14
Question 13	15
Question 14	16
Question 15	18
Question 16	19
Question 17	20
Question 18	21
Question 19	22
Question 20	23

Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**Preparing for the test**’ section on the Admissions Testing website.

Question 1

Approach 1: using the equations of the circles

We can find the equation by solving the equations of the circles simultaneously.

The equations are

$$\begin{aligned}(x+2)^2 + (y-1)^2 &= r^2 \\ (x-3)^2 + (y+2)^2 &= r^2\end{aligned}$$

(We do not know the radius, so we have called it r ; we are given that both circles have the same radius.)

We can expand the brackets to give

$$\begin{aligned}x^2 + 4x + 4 + y^2 - 2y + 1 &= r^2 \\ x^2 - 6x + 9 + y^2 + 4y + 4 &= r^2\end{aligned}$$

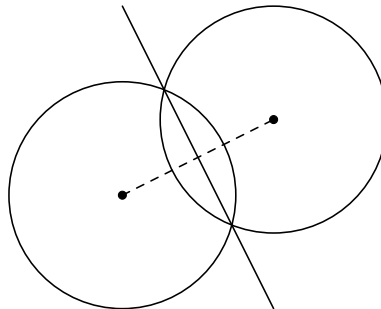
Subtracting these equations then gives an equation which must be satisfied by any points lying on both circles:

$$10x - 6y - 8 = 0$$

which simplifies to $5x - 3y = 4$, and so the answer is F.

Approach 2: using geometry

The circles have the same radius, so by symmetry, the straight line through the two points of intersection must be the perpendicular bisector of the line segment joining the two circle centres, as shown in this diagram:



In our case, the two centres are at $(-2, 1)$ and $(3, -2)$, so the midpoint is at $(\frac{1}{2}, -\frac{1}{2})$. The gradient of the line joining the centres is $\frac{-2-1}{3-(-2)} = -\frac{3}{5}$. Therefore the gradient of the perpendicular bisector is $\frac{5}{3}$, and it has equation

$$y - (-\frac{1}{2}) = \frac{5}{3}(x - \frac{1}{2})$$

Multiplying by 3 gives

$$3y + \frac{3}{2} = 5(x - \frac{1}{2})$$

which rearranges to give

$$5x - 3y = 4$$

which is option F.

Question 2

We first find the turning points by differentiation. We have

$$\frac{dy}{dx} = 3x^2 - 6 = 0 \text{ if and only if } x^2 = 2$$

so $\alpha = -\sqrt{2}$ and $\beta = \sqrt{2}$.

We can now compute the integral:

$$\begin{aligned} \int_{\alpha}^{\beta} x^3 - 6x + 3 \, dx &= \left[\frac{1}{4}x^4 - 3x^2 + 3x \right]_{-\sqrt{2}}^{\sqrt{2}} \\ &= \left(\frac{1}{4}(\sqrt{2})^4 - 3(\sqrt{2})^2 + 3(\sqrt{2}) \right) - \left(\frac{1}{4}(-\sqrt{2})^4 - 3(-\sqrt{2})^2 + 3(-\sqrt{2}) \right) \\ &= 6\sqrt{2} \end{aligned}$$

where to get the last line, we note that the first terms in each bracket cancel with each other, and similarly for the second terms, so we are left with just $3\sqrt{2} - (-3\sqrt{2})$. This is option F.

Question 3

Let us write the arithmetic progression as

$$\frac{1}{2}, \quad \frac{1}{2} + d, \quad \frac{1}{2} + 2d, \quad \frac{1}{2} + 3d, \quad \dots$$

and the geometric progression as

$$\frac{1}{2}, \quad \frac{1}{2}r, \quad \frac{1}{2}r^2, \quad \frac{1}{2}r^3, \quad \dots$$

Then the given information about the second terms and third terms can be written algebraically as

$$\frac{1}{2} + d + \frac{1}{2}r = \frac{1}{2}$$

$$\frac{1}{2} + 2d + \frac{1}{2}r^2 = \frac{1}{8}$$

The first equation simplifies to give $d + \frac{1}{2}r = 0$, so $d = -\frac{1}{2}r$. We can then substitute this into the second equation to give

$$\frac{1}{2} + \frac{1}{2}r^2 - r = \frac{1}{8}$$

(We could equivalently have eliminated d by subtracting twice the first equation from the second.) Multiplying by 8 and rearranging slightly gives

$$4r^2 - 8r + 3 = 0.$$

We can solve this by factorising or using the formula, for example; this gives

$$r = \frac{8 \pm \sqrt{(-8)^2 - 4 \times 4 \times 3}}{2 \times 4} = \frac{1}{2} \text{ or } \frac{3}{2}$$

Since the geometric progression is convergent, we must have $|r| < 1$, so $r = \frac{1}{2}$ and the sum to infinity is

$$S_{\infty} = \frac{a}{1-r} = \frac{\left(\frac{1}{2}\right)}{1-\frac{1}{2}} = 1$$

so the answer is option G.

Question 4

This function is a quadratic in $u = 2^x$; we can rewrite it as

$$u^2 - 2^3u + 4$$

or

$$u^2 - 8u + 4.$$

We can complete the square to express this as

$$(u - 4)^2 - 4^2 + 4 = (u - 4)^2 - 12.$$

The minimum therefore occurs when $u = 4$, at which point the function value is 12.

We must also check that this value can actually be attained: $u = 4$ if and only if $2^x = 4$, which happens when $x = 2$, so this minimum is attained, and the correct answer is B.

The minimum can also be found using calculus: differentiating $u^2 - 8u + 4$ gives $2u - 8$; this is zero when $u = 4$, at which point the function value is $4^2 - 8 \times 4 + 4 = -12$; this is a minimum as the function is a quadratic with positive u^2 coefficient.

Question 5

We can rewrite $f(9)$, $f(16)$ and $f(24)$ using the information given about the function:

$$\begin{aligned}
 f(9) &= f(3 \times 3) \\
 &= f(3)f(3) && \text{as 9 is a multiple of 3} \\
 f(16) &= f(16 \times 1) \\
 &= 16 && \text{as 16 is not a multiple of 3} \\
 f(24) &= f(3 \times 8) \\
 &= f(3)f(8) && \text{as 24 is a multiple of 3} \\
 &= f(3)f(8 \times 1) \\
 &= f(3) \times 8 && \text{as 8 is not a multiple of 3} \\
 &= 8f(3)
 \end{aligned}$$

Substituting these into the given equation then gives

$$(f(3))^2 + 16 - 8f(3) = 0$$

which is a quadratic in $f(3)$; we can factorise it to give

$$(f(3) - 4)^2 = 0$$

which has one (repeated) solution: $f(3) = 4$. Hence the answer is option F.

Question 6

This is almost a function in $\cos x$ except for the $\sin^2 x$ term, so we rewrite this in terms of $\cos x$:

$$f(x) = \frac{\cos x + 3}{7 + 5 \cos x - (1 - \cos^2 x)} = \frac{\cos x + 3}{6 + 5 \cos x + \cos^2 x}$$

We can then factorise the denominator to give

$$f(x) = \frac{\cos x + 3}{(\cos x + 2)(\cos x + 3)}$$

and the $\cos x + 3$ factor cancels. (We should also check that this can never be zero; if it were, the function would be undefined at some points.) Therefore we can write

$$f(x) = \frac{1}{\cos x + 2}$$

The maximum value of this function occurs where $\cos x = -1$, giving $f(x) = 1$, and the minimum value occurs where $\cos x = +1$, giving $f(x) = \frac{1}{3}$. The difference between the minimum and maximum values is therefore $1 - \frac{1}{3} = \frac{2}{3}$, which is option D.

Question 7

Let us start with the condition $xf(x) > 0$ for all non-zero values of x . When $x > 0$, this means that $f(x) > 0$, as we can divide the inequality by x and the direction of the inequality is unchanged. On the other hand, when $x < 0$, this gives $f(x) < 0$, since dividing by x now changes the direction of the inequality. In summary:

$$\begin{aligned} f(x) &> 0 \text{ when } x > 0 \\ f(x) &= 0 \text{ when } x = 0 \\ f(x) &< 0 \text{ when } x < 0 \end{aligned}$$

Now let us turn to the integrals. The final question asks about an integral from -2 to 0 , so we would do well to split the two given integrals into the ranges -2 to 0 and 0 to 2 :

$$\begin{aligned} \int_{-2}^2 f(x) \, dx &= \int_{-2}^0 f(x) \, dx + \int_0^2 f(x) \, dx \\ &= 4 \\ \int_{-2}^2 |f(x)| \, dx &= \int_{-2}^0 |f(x)| \, dx + \int_0^2 |f(x)| \, dx \\ &= \int_{-2}^0 -f(x) \, dx + \int_0^2 f(x) \, dx \\ &= -\int_{-2}^0 f(x) \, dx + \int_0^2 f(x) \, dx \\ &= 8 \end{aligned}$$

Therefore

$$\begin{aligned} \int_{-2}^0 f(x) \, dx + \int_0^2 f(x) \, dx &= 4 \\ -\int_{-2}^0 f(x) \, dx + \int_0^2 f(x) \, dx &= 8 \end{aligned}$$

If we now add these two simultaneous equations together, the $\int_{-2}^0 f(x) \, dx$ terms cancel and we are left with $2 \int_0^2 f(x) \, dx = 12$, so $\int_0^2 f(x) \, dx = 6$ and $\int_{-2}^0 f(x) \, dx = -2$.

The question asks us to evaluate $\int_{-2}^0 f(|x|) \, dx$. Thinking about graph transformations, the graph of $y = f(x)$ is transformed into the graph of $y = f(|x|)$ by removing the left half of the graph (left of the y -axis) and reflecting the right half of the graph in the y -axis (the line $x = 0$). The integral of this transformed function from -2 to 0 is therefore equal to the integral of the original function from 0 to 2 , thus

$$\int_{-2}^0 f(|x|) \, dx = \int_0^2 f(x) \, dx = 6$$

and the correct option is G.

Question 8

A straight line meets a quadratic at exactly one point if and only if there is a repeated root to the simultaneous equations. Substituting for y in the first pair gives

$$2x + 3 = x^2 + bx + c.$$

This rearranges to $x^2 + (b - 2)x + (c - 3) = 0$. As this has a repeated root, its discriminant must be zero, so

$$(b - 2)^2 - 4(c - 3) = 0.$$

The second line likewise gives $(b - 4)^2 - 4(c + 2) = 0$. Expanding and simplifying these two equations gives the following simultaneous equations for b and c :

$$b^2 - 4b - 4c = -16$$

$$b^2 - 8b - 4c = -8$$

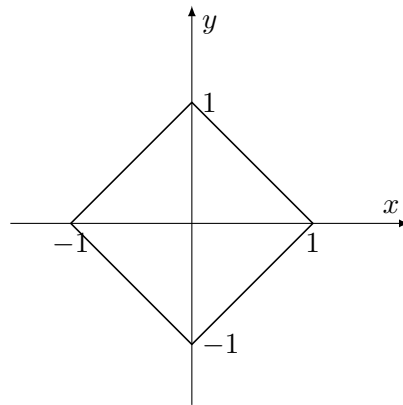
Subtracting these two equations gives $4b = -8$ so $b = -2$ and hence $c = 7$. Therefore $b - c = -9$, which is option A.

Question 9

We first need to work out what the graph looks like. There are four cases for x and y depending on whether there are positive or negative. We list them in a table:

quadrant	x	y	$ x $	$ y $	equation	rearranged to $y = \dots$
1st	+	+	x	y	$x + y = 1$	$y = -x + 1$
2nd	-	+	$-x$	y	$-x + y = 1$	$y = x + 1$
3rd	-	-	$-x$	$-y$	$-x - y = 1$	$y = -x - 1$
4th	+	-	x	$-y$	$x - y = 1$	$y = x - 1$

We therefore have (at most) four lines making up this graph, and we can sketch these. We need to recall that we only draw the part of each line that appears within the relevant quadrant; for example, $y = x + 1$ only appears in the 2nd quadrant.



This shape is a square. We can think of it as made up of 4 right angled triangles, each with base and height equal to 1, so each has area $\frac{1}{2} \times 1 \times 1 = \frac{1}{2}$, and hence the square has area 2. Thus the correct option is C.

Question 10

The interval has width $2 - \frac{1}{2} = \frac{3}{2}$, so each strip will have width $\frac{1}{2}$. The boundaries of the three strips will be at $\frac{1}{2}$, 1, $\frac{3}{2}$ and 2. The trapezium rule then gives the estimate (writing $f(x) = 2 \log_{10} x$):

$$\int_{\frac{1}{2}}^2 f(x) \, dx \approx \frac{1}{2} \times \frac{1}{2} (f(\frac{1}{2}) + 2f(1) + 2f(\frac{3}{2}) + f(2))$$

We can now work out the individual function values:

$$\begin{aligned} f(\frac{1}{2}) &= 2 \log_{10} \frac{1}{2} = -2 \log_{10} 2 \\ f(1) &= 2 \log_{10} 1 = 0 \\ f(\frac{3}{2}) &= 2 \log_{10} \frac{3}{2} = 2 \log_{10} 3 - 2 \log_{10} 2 \\ f(2) &= 2 \log_{10} 2 \end{aligned}$$

Therefore we get the approximation

$$\begin{aligned} \int_{\frac{1}{2}}^2 f(x) \, dx &\approx \frac{1}{2} \times \frac{1}{2} (-2 \log_{10} 2 + 2 \times 0 + 2(2 \log_{10} 3 - 2 \log_{10} 2) + 2 \log_{10} 2) \\ &= \frac{1}{4} (4 \log_{10} 3 - 4 \log_{10} 2) \\ &= \log_{10} 3 - \log_{10} 2 \\ &= \log_{10} \frac{3}{2} \end{aligned}$$

which is option B.

Question 11

The function is decreasing when $f'(x) < 0$ (and possibly also at points where $f'(x) = 0$). We have

$$f(x) = x^{15/7} - x^{8/7} + x^{1/7}$$

so

$$\begin{aligned} f'(x) &= \frac{15}{7}x^{8/7} - \frac{8}{7}x^{1/7} + \frac{1}{7}x^{-6/7} \\ &= \frac{1}{7}x^{-6/7}(15x^2 - 8x + 1) \\ &= \frac{1}{7}x^{-6/7}(5x - 1)(3x - 1) \end{aligned}$$

This is negative when $(5x - 1)(3x - 1) < 0$ (note that we are only considering the region $x > 0$ here).

We note that $(5x - 1)(3x - 1) = 0$ exactly when $x = \frac{1}{5}$ or $x = \frac{1}{3}$ and is therefore negative when $\frac{1}{5} < x < \frac{1}{3}$. Therefore the function $f(x)$ is decreasing when $\frac{1}{5} < x < \frac{1}{3}$ (and possibly also at the endpoints of this interval).

Thus the fraction of the interval $0 < x < 1$ for which $f(x)$ is decreasing is $\frac{1}{3} - \frac{1}{5} = \frac{2}{15}$, which is option A.

Question 12

We can complete the square with the first function: we have

$$x^4 - p^2x^2 = (x^2 - \frac{1}{2}p^2)^2 - \frac{1}{4}p^4$$

This attains its minimum value when the parenthesised expression is 0, that is when $x = \pm \frac{1}{\sqrt{2}}p$. At that point, $x^4 - p^2x^2 = -\frac{1}{4}p^4 = -9$, so $p^4 = 36$. Since p is a real number, we must have $p^2 = 6$ (rather than -6).

We now consider $x^2 - px + 6$. We can again complete the square to get

$$x^2 - px + 6 = (x - \frac{1}{2}p)^2 - \frac{1}{4}p^2 + 6.$$

The minimum of this function again occurs when the parenthesised expression is 0, that is when $x = \frac{1}{2}p$. At this point,

$$x^2 - px + 6 = -\frac{1}{4}p^2 + 6 = -\frac{1}{4} \times 6 + 6 = \frac{9}{2}$$

and so the answer is option E.

Alternatively, we could have used calculus instead of completing the square to find the minima; the results would be the same.

Question 13

Let us first work out the value of the integral in the summation (the summand). We have

$$\begin{aligned}\int_0^r f(x) \, dx &= \int_0^1 f(x) \, dx + \int_1^2 f(x) \, dx + \cdots + \int_{r-1}^r f(x) \, dx \\ &= (0 + 1) + (1 + 1) + \cdots + ((r - 1) + 1) \\ &= 1 + 2 + \cdots + r \\ &= \frac{1}{2}r(r + 1).\end{aligned}$$

We are thus being asked to evaluate

$$\sum_{r=1}^8 \frac{1}{2}r(r + 1).$$

There are algebraic methods to do this (as mentioned below), but they are not on the specification. The most straightforward approach here is to just perform the calculation manually:

$$\sum_{r=1}^8 \frac{1}{2}r(r + 1) = 1 + 3 + 6 + 10 + 15 + 21 + 28 + 36 = 120,$$

which is option C.

In general, it turns out that

$$\sum_{r=1}^n \frac{1}{2}r(r + 1) = \frac{1}{6}n(n + 1)(n + 2).$$

The expressions $r(r + 1)$ and $n(n + 1)(n + 2)$ are known as *rising powers*, and can be written as $r^{\overline{2}}$ and $n^{\overline{3}}$ respectively. On multiplying by 2, the formula then becomes

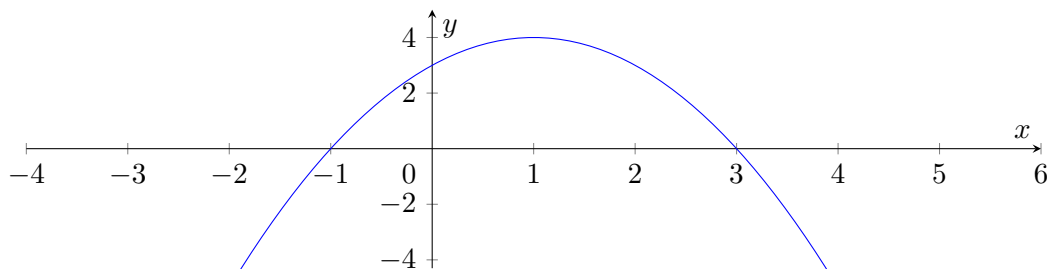
$$\sum_{r=1}^n r^{\overline{2}} = \frac{1}{3}n^{\overline{3}},$$

which looks very similar to the calculus formula $\int_0^x t^2 \, dt = \frac{1}{3}x^3$. There is a much longer discussion of sums such as this in Graham, Knuth and Patashnik, *Concrete Mathematics*.

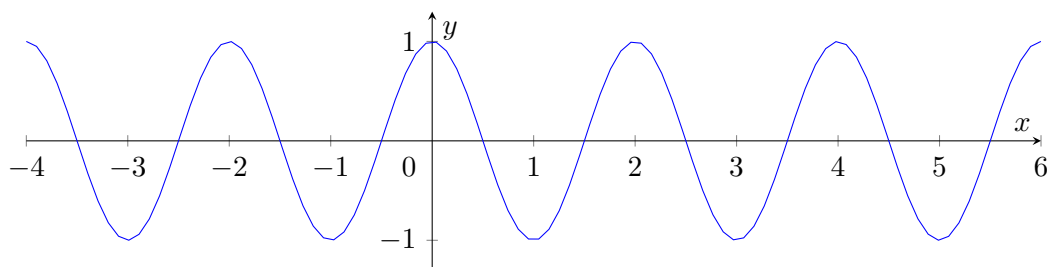
Question 14

We do not know a method of solving equations involving both trigonometric functions (on the right-hand side) and quadratics (on the left-hand side), but the question only asks us to count the number of solutions, not to solve the equation. So we will instead sketch the graphs of the left-hand and right-hand sides and see how many times they intersect.

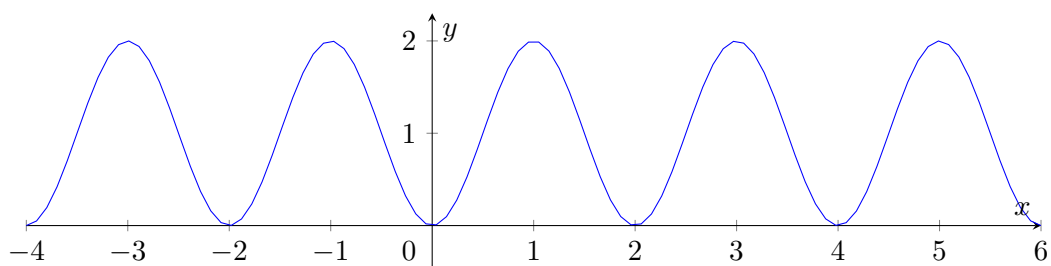
The graph of $y = (x+1)(3-x)$ is a quadratic with roots at $x = -1$ and $x = 3$. When expanded, it becomes $y = -x^2 + \dots$, so it has a maximum point midway between the roots, at $x = 1$, $y = 4$.



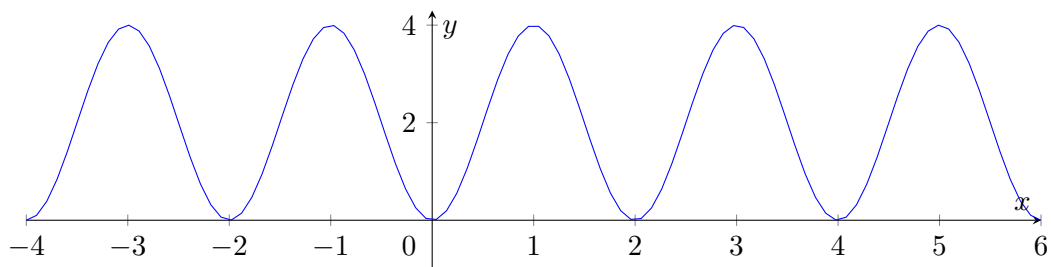
The graph of $y = 2(1 - \cos(\pi x))$ is a transformed cosine graph. $y = \cos(\pi x)$ has a period of 2: it is the graph of $y = \cos x$ scaled by a factor of $1/\pi$ in the x -direction, and looks like this:



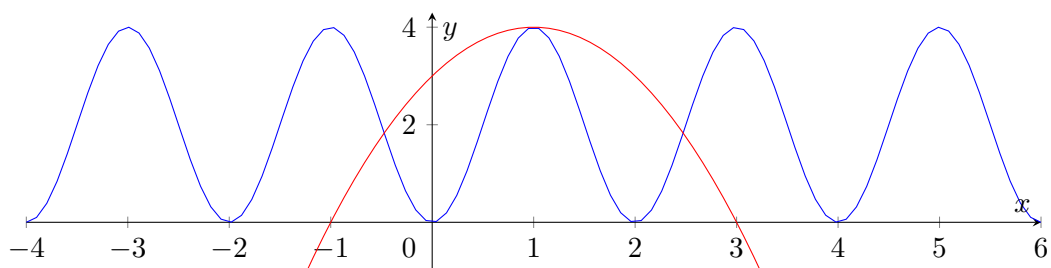
We can then work out the graph of $y = 1 - \cos(\pi x)$ by calculating the values of y at a few key points, or by observing that the graph of $y = -\cos(\pi x)$ is the reflection of $y = \cos(\pi x)$ in the x -axis, and then $y = 1 - \cos(\pi x)$ is this translated by 1 unit in the y -direction. The result is:



Finally, the graph of $y = 2(1 - \cos(\pi x))$ is the previous graph scaled by a factor of 2 in the y -direction:

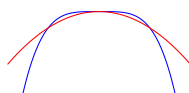


We can now superimpose the graph of $y = (x + 1)(3 - x)$ on top of $y = 2(1 - \cos(\pi x))$. We see that the trigonometric graph is never below the x -axis, so the only possible points of intersection are between $x = -1$ and $x = 3$. There is one point of intersection at $(1, 4)$, one between $x = -1$ and $x = 0$ and one between $x = 2$ and $x = 3$, so there are three points of intersection in total.



Thus the correct answer is option B.

You might wonder about a subtle point: we see that the two graphs touch each other at $(1, 4)$, but how do we know that the trigonometric graph is ‘sharper’ there, so does not also cross the quadratic graph close to $x = 1$, something like this:



This is a plausible possibility, but it turns out not to be the case for reasons that are beyond the scope of the specification.

Question 15

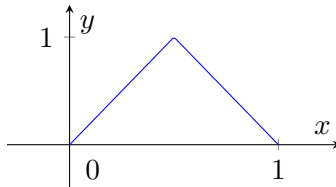
We can write the integral out in full by substituting the value of $g(x)$:

$$\begin{aligned}\int_0^1 g(x) \, dx &= \int_0^1 \sum_{r=1}^{10} f(2^{r-1}x) \, dx \\ &= \int_0^1 f(2^0x) + f(2^1x) + f(2^2x) + \cdots + f(2^9x) \, dx \\ &= \int_0^1 f(2^0x) \, dx + \int_0^1 f(2^1x) \, dx + \int_0^1 f(2^2x) \, dx + \cdots + \int_0^1 f(2^9x) \, dx\end{aligned}$$

So we can calculate the integral of $g(x)$ if we can calculate the integral of each $f(2^{r-1}x)$.

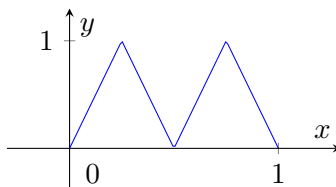
Now $\int_0^1 f(2^0x) \, dx = \int_0^1 f(x) \, dx$ is just the area under the graph of $y = f(x)$ between $x = 0$ and $x = 1$, which is $\frac{1}{2}$ (it is a right angled triangle with base and height both 1).

To work out $\int_0^1 f(2x) \, dx$, we consider the graph of $y = f(2x)$. This is the graph of $y = f(x)$ scaled by a factor of $1/2$ in the x -direction, so looks like this:

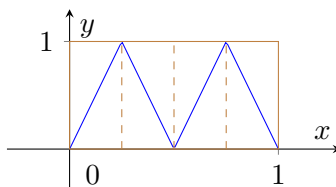


The area between $x = 0$ and $x = 1$ is again $\frac{1}{2}$.

The graph of $y = f(2^2x)$ between $x = 0$ and $x = 1$ consists of two triangles:



Helpfully, the area under this graph is still $\frac{1}{2}$: this is perhaps easiest to see by noticing that we can draw a rectangle (actually a square) around the graph:



In each rectangular slice, half of the area is below the graph and half is above, so the total area below the graph is $\frac{1}{2}$.

This same reasoning will apply for each of the functions $f(2^{r-1}x)$, and so the total area is $10 \times \frac{1}{2} = 5$, which is option C.

Question 16

We can expand the binomial expression in ascending powers of x using the binomial theorem:

$$(a + bx)^n = a^n + \binom{n}{1}a^{n-1}bx + \binom{n}{2}a^{n-2}b^2x^2 + \binom{n}{3}a^{n-3}b^3x^3 + \cdots + \binom{n}{n-1}ab^{n-1}x^{n-1} + b^nx^n$$

The third term in ascending powers of x is $\binom{n}{2}a^{n-2}b^2x^2 = 105x^2$.

The fourth term in ascending powers of x is $\binom{n}{3}a^{n-3}b^3x^3 = 210x^3$.

The fourth term in descending powers of x is $\binom{n}{n-3}a^3b^{n-3}x^{n-3} = 210x^3$.

From the last statement, we see that $n = 6$. Then the first two statements give:

$$\begin{aligned}\binom{6}{2}a^4b^2 &= 105 \\ \binom{6}{3}a^3b^3 &= 210\end{aligned}$$

Now $\binom{6}{2} = 15$ and $\binom{6}{3} = 20$, so

$$\begin{aligned}a^4b^2 &= 7 \\ a^3b^3 &= \frac{21}{2}\end{aligned}$$

Dividing these two equations gives

$$\frac{a}{b} = \frac{2}{3}$$

and so $\left(\frac{a}{b}\right)^2 = \frac{4}{9}$, which is option B.

Question 17

We note that $x^2 + y^2 \geq 0$ for all x and y , so we need only consider non-negative values of $x^2 + y^2$ that satisfy this equation. Those with $x^2 + y^2 \leq 8\pi$ are

$$x^2 + y^2 = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \frac{37\pi}{6}, \frac{41\pi}{6}$$

(These were obtained by noting that the solutions with $x^2 + y^2 \leq 2\pi$ are $\frac{\pi}{6}$ and $\frac{5\pi}{6}$, and then adding multiples of $2\pi = \frac{12\pi}{6}$ to these.)

Consider the first value; $x^2 + y^2 = \frac{\pi}{6}$; this is a circle with radius $\sqrt{\pi/6}$. Likewise, the other circles have radii $\sqrt{5\pi/6}, \dots, \sqrt{41\pi/6}$. Taking out a common factor of $\sqrt{\pi/6}$, we see that the circles have radii:

$$\rho\sqrt{\frac{\pi}{6}}, \quad \rho = 1, \sqrt{5}, \sqrt{13}, \sqrt{17}, \sqrt{25}, \sqrt{29}, \sqrt{37}, \sqrt{41}$$

We can approximate these square roots to get an idea of how they grow:

$$\rho \approx 1, 2, 3.5, 4, 5, 5.5, 6, 6.5$$

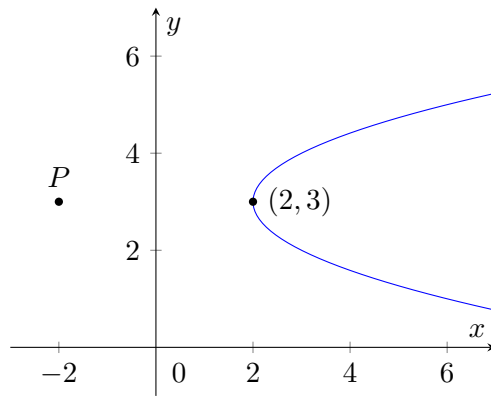
So the smallest three circles are quite far apart, but after that, they get closer and closer, which is graph A. (We can be more precise than this, and observe that the circles appear in close pairs with larger gaps between the pairs, but that is not needed to answer this question.)

Question 18

We should start by sketching the original curve. This is a quadratic in y . It does not factorise, but we can complete the square:

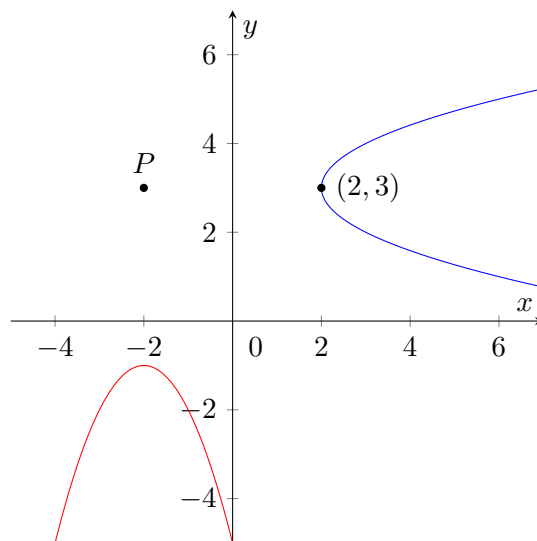
$$x = (y - 3)^2 + 2$$

It therefore has its vertex at $y = 3$, $x = 2$, and $x \geq 2$ for all values of y . We can then sketch it:



The vertex of the parabola is 4 units to the right of P .

Now rotating this by 90° clockwise about P moves the vertex to 4 units below P , that is $(-2, -1)$, giving the following:



This curve then has equation $y = -(x + 2)^2 - 1$, which we can expand to $y = -x^2 - 4x - 5$ giving option B.

Question 19

We can start by solving the equation before considering the possible values of x .

We have

$$\sin(4^{\cos \theta} \times 60^\circ) = \pm \frac{\sqrt{3}}{2}$$

so that

$$4^{\cos \theta} \times 60^\circ = 60^\circ, 120^\circ, 240^\circ, 300^\circ, \dots$$

(noting that the left hand side is always positive, so we do not need to consider negative values for the right hand side).

Dividing by 60° gives

$$4^{\cos \theta} = 1, 2, 4, 5, \dots$$

and so

$$\cos \theta = 0, \frac{1}{2} \text{ or } 1;$$

there are no values of $\cos \theta$ for which $4^{\cos \theta} > 4$.

We can now find the possible values of θ :

- $\cos \theta = 0$ gives $\theta = 90^\circ, 270^\circ, 450^\circ, \dots$
- $\cos \theta = \frac{1}{2}$ gives $\theta = 60^\circ, 300^\circ, 420^\circ, \dots$
- $\cos \theta = 1$ gives $\theta = 0^\circ, 360^\circ, \dots$

Collecting these solutions and arranging them in increasing order gives $\theta = 0^\circ, 60^\circ, 90^\circ, 270^\circ, 300^\circ, \dots$

We are told that there are exactly three solutions in the range $0^\circ \leq \theta \leq x^\circ$. Therefore $x \geq 90$ to get the third solution, and $x < 270$ to avoid getting the fourth solution. This gives the answer $90 \leq x < 270$, which is option B.

Question 20

We can combine the logarithms on the right and rearrange the left hand side to give

$$\log_{10}(x - y)^2 = \log_{10}((2 - 2x)(y + 5))$$

and so

$$(x - y)^2 = (2 - 2x)(y + 5).$$

Expanding the brackets gives

$$x^2 - 2xy + y^2 = 2y + 10 - 2xy - 10x$$

which rearranges to

$$x^2 + y^2 + 10x - 2y = 10.$$

Completing the squares gives

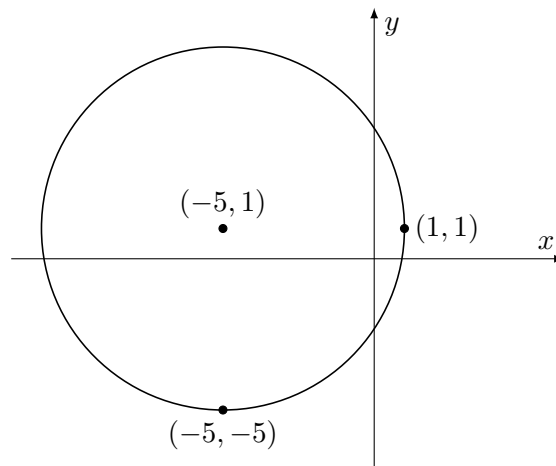
$$(x + 5)^2 + (y - 1)^2 = 36$$

which is a circle with centre $(-5, 1)$ and radius 6.

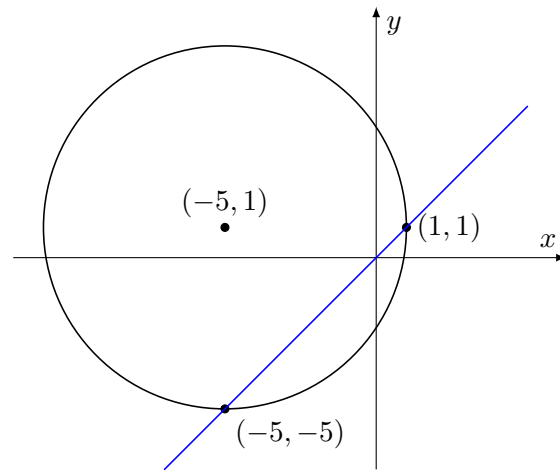
The obvious answer is therefore 12π , but we must take care: does every point on the circle satisfy the original equation? The manipulations we have performed make assumptions; in particular, for the logarithms to be defined, we require $x - y > 0$, $2 - 2x > 0$ and $y + 5 > 0$, which we can rearrange to

$$x > y, \quad x < 1 \quad \text{and} \quad y > -5$$

Let us sketch the circle and these three inequalities. Here is the circle with two points on the circumference marked:



The restriction $x < 1$ holds for all of the points on the circle except for the single point $(1, 1)$, and the restriction $y > -5$ holds everywhere except for $(-5, -5)$. The restriction $x > y$ is more significant for us. We can draw the line $x = y$ on our sketch:



The region $x > y$ is below this line, and we see that only part of the circle is included in the region. Since the points of intersection of the line $x = y$ with the circle are at $(1, 1)$ and $(-5, -5)$, it is clear that exactly one quarter of the circle lies in the region $x > y$. Therefore the length of the curve with the given equation is $\frac{1}{4}$ of the circumference of the circle, that is, $\frac{1}{4} \times 12\pi = 3\pi$, and the correct option is D.

This document was initially designed for print and as such does not reach accessibility standard WCAG 2.1 in a number of ways including missing text alternatives and missing document structure.

If you need this document in a different format please email admissionstesting@cambridgeassessment.org.uk telling us your name, email address and requirements and we will respond within 15 working days.

This document contains maths equations and symbols which at present we are not able to produce adequately for accessibility. We are working to improve this.

We are Cambridge Assessment Admissions Testing, part of the University of Cambridge. Our research-based tests provide a fair measure of skills and aptitude to help you make informed decisions. As a trusted partner, we work closely with universities, governments and employers to enhance their selection processes.

Cambridge Assessment
Admissions Testing
The Triangle Building
Shaftesbury Road
Cambridge
CB2 8EA
United Kingdom

Admissions tests support: admissionstesting.org/help

Test of Mathematics for University Admission, 2021 Paper 2

Worked Solutions

Version 1.0, March 2022

Contents

Introduction for students	2
Question 1	3
Question 2	4
Question 3	5
Question 4	6
Question 5	7
Question 6	8
Question 7	10
Question 8	11
Question 9	12
Question 10	13
Question 11	14
Question 12	15
Question 13	16
Question 14	17
Question 15	18
Question 16	19
Question 17	21
Question 18	22
Question 19	24
Question 20	25

Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**Preparing for the test**’ section on the Admissions Testing website.

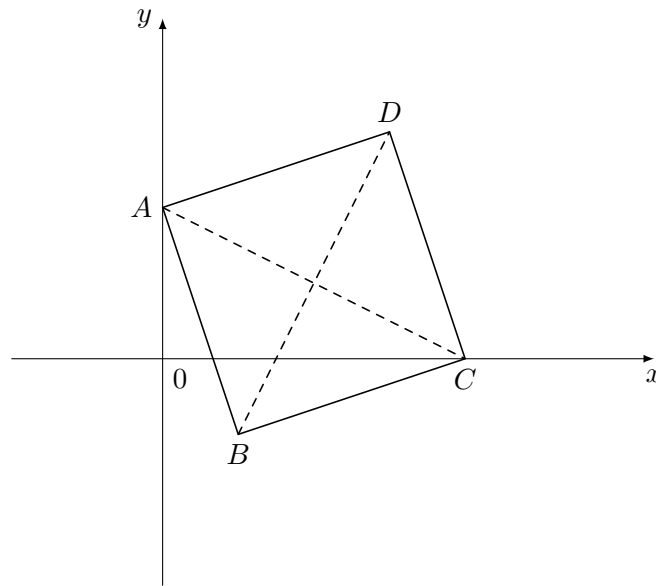
Question 1

We write the integral as powers of x and then integrate:

$$\begin{aligned}
 \int_1^4 \left(3\sqrt{x} + \frac{4}{x^2} \right) dx &= \int_1^4 3x^{\frac{1}{2}} + 4x^{-2} dx \\
 &= \left[\frac{3x^{\frac{3}{2}}}{(\frac{3}{2})} + \frac{4x^{-1}}{-1} \right]_1^4 \\
 &= \left[2x^{\frac{3}{2}} - 4x^{-1} \right]_1^4 \\
 &= \left(2 \times 8 - 4 \times \frac{1}{4} \right) - \left(2 \times 1 - 4 \times 1 \right) \\
 &= 15 - (-2) \\
 &= 17
 \end{aligned}$$

Question 2

We can sketch the situation:



We know that the two diagonals of the square bisect each other at right angles, so this question requires us to find the perpendicular bisector of AC .

The midpoint of AC is $(\frac{0+4}{2}, \frac{2+0}{2}) = (2, 1)$, which is the centre of the square.

The gradient of AC is $\frac{0-2}{4-0} = -\frac{1}{2}$, so the gradient of BD is 2.

The equation of BD is therefore $y - 1 = 2(x - 2)$, which rearranges to give $y = 2x - 3$, which is option E.

Question 3

The probability is given as a fraction in its lowest terms.

The probability that a student does not wear glasses is $\frac{11}{15}$, so the ratio of glasses to non-glasses is 4 : 11, in its lowest terms.

Therefore the number of glasses/non-glasses wearers could be 4 and 11 (total 15), or 8 and 22 (total 30), or ...

Necessary means that it follows as a consequence.

Therefore I and III are not necessary (there could be 22 non-glasses wearers or the class could contain 15 students).

The total must be a multiple of 15, and so the number of students is divisible by 3.

The correct answer is therefore C.

Question 4

- I $bc = 200$, a is a factor of 200, a is a factor of b , so not a counterexample.
- II $bc = 16$, a is a factor of 16, a is not a factor of b and a is not a factor of c , so this is a counterexample.
- III $bc = 84$, a is a factor of 84, a is not a factor of b but a is a factor of c , so not a counterexample.

The correct option is therefore C.

Question 5

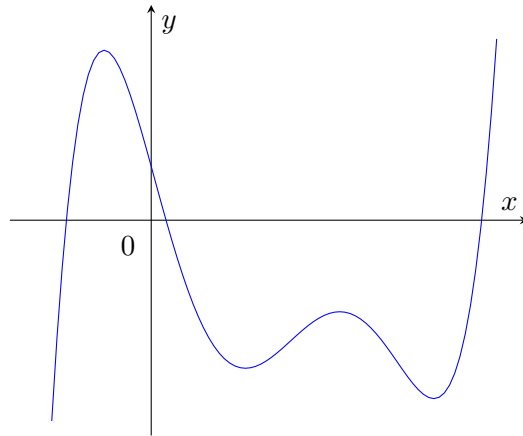
Line A is a known identity, so this is fine.

Line B rearranges the identity to get $\cos^2 x = 1 - \sin^2 x$ and then takes the square root of both sides. However, when we take the square root, we always get the positive (or zero) root, but $\cos x$ is negative for some values of x . So this is where the error is introduced.

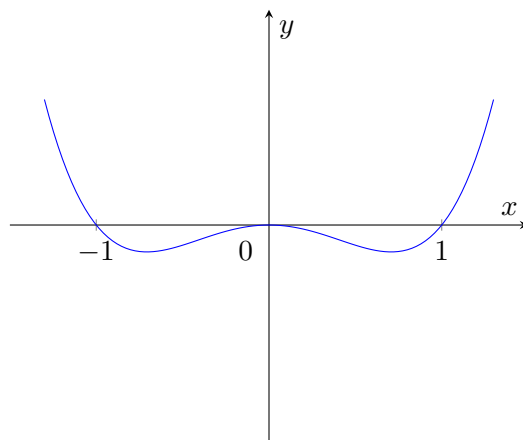
Another way of seeing this is to start with the nonsensical $0 = 4$ of line E, and substitute $x = \pi$ in each line to find the first one which fails; it is, indeed, line B, as this gives $-1 = 1$.

Question 6

If $f(x) = 0$ has exactly three roots then there must be a stationary point between the first and second root, and another stationary point between the second and third roots, but there may be other stationary points. For example, here is a sketch of a degree 5 polynomial $y = f(x)$ which has $f(x) = 0$ for three real values of x but $f'(x) = 0$ for five real values of x :

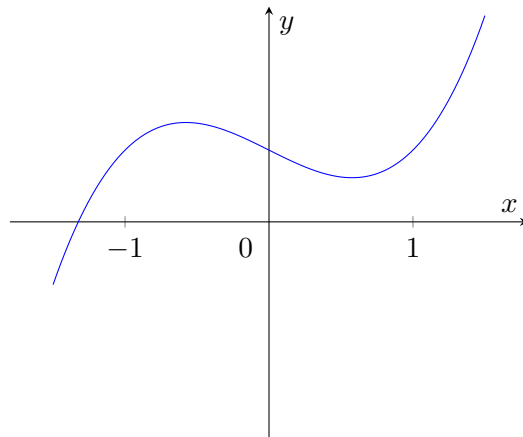


A second example is $f(x) = x^4 - x^2$. Here, the graph of $y = f(x)$ touches the x -axis at $x = 0$, so $f(x) = 0$ has exactly three real roots $x = -1, 0$, and 1 , but three stationary points $x = \pm \frac{1}{\sqrt{2}}$ and $x = 0$:



So P is not sufficient for Q .

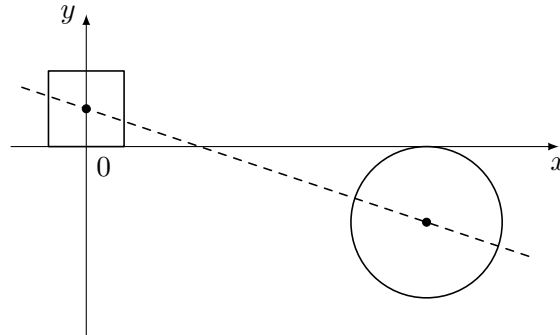
It is also not necessary, for example $f(x) = x^3 - x + 1$ has exactly two stationary points but only one real root:



So option D is correct.

Question 7

We start by drawing the square and the circle. Since the straight line bisects the area of each shape, it must pass through their centres, as shown.



This straight line therefore passes through the centre of the square at $(0, 1)$ and the centre of the circle at $(9, -2)$. The gradient of this line is then $\frac{-2-1}{9-0} = -\frac{1}{3}$, so the line has equation $y = -\frac{1}{3}x + 1$.

The line meets the x -axis where $y = 0$, so $0 = -\frac{1}{3}x + 1$, giving $x = 3$.

Hence the correct answer is B.

Question 8

If $p(a) = p(b) = c$, say, then if the graph of $y = p(x)$ (strictly) increases as it leaves (a, c) towards $x = b$, but then it must decrease again to get back to (b, c) , so there must be a stationary point between $x = a$ and $x = b$. Similarly, if the graph (strictly) decreases as it leaves (a, c) , it must then increase again, which also results in a stationary point. The only other possibility is that the graph of $y = p(x)$ is constant, in which case $p'(x) = 0$ throughout. In all cases, $(*)$ holds, so $p(a) = p(b)$ is sufficient for $(*)$.

However, it is not necessary for $(*)$: if $p(x) = x^2$ and $a = -1$, $b = 2$, then this satisfies $(*)$ (with $c = 0$) but $p(a) \neq p(b)$.

Thus the correct option is C.

Question 9

- I Every cubic crosses the x -axis, so this condition is sufficient for $f(x) = 0$ to have a real root. This also follows from III below if we take u to be a very large positive number and v to be a very large negative number.
- II Consider, for example, $f(x) = x^2 + 1$. This quadratic has $f'(x) = 2x$ so $f'(0) = 0$, but there is no real x for which $x^2 + 1 = 0$. So this condition is not sufficient.
- III As $f(u)f(v) < 0$, we must have $f(u) < 0$ and $f(v) > 0$ or the other way round; either way, the graph of $y = f(x)$ has values above and below the x -axis. Since $f(x)$ is a polynomial, it must therefore cross the x -axis, and so the equation $f(x) = 0$ has a real root. (If we did not know that $f(x)$ was a polynomial, this would not be sufficient; consider $f(x) = \frac{1}{x}$ for example.)

The correct option is therefore C.

Question 10

To be a counterexample, we need the antecedent (the ‘if’ part) to be true but the consequent (the ‘then’ part) to be false.

n	n is prime	a_n multiple of 3 or 5
1	false	–
2	true	true
3	true	true
4	false	–
5	true	false
6	false	–
7	true	false

Therefore $n = 5$ and $n = 7$ are both counterexamples, and the smallest counterexample is $n = 5$, which is option E.

Question 11

We start by checking that each step works.

Line (II) follows from line (I) by expanding the brackets.

The algebraic expression in line (III) expands as $x^2 + 2xy + y^2 - 4xy$, so it follows from line (II).

Line (IV) follows from line (III) by multiplying by x^2y^2 . Since $x^2y^2 \geq 0$, the inequality keeps the same direction, so this is a valid deduction.

However, how does line (V) follow from line (IV)? It assumes that $a = xy(x + y)$ and $b = xy$ in order to make the deduction.

This argument has therefore shown that *if* $a = xy(x + y)$ and $b = xy$, *then* $a^2 - 4b^3 \geq 0$, or even more explicitly: *if* there exist real numbers x and y such that $a = xy(x + y)$ and $b = xy$, *then* $a^2 - 4b^3 \geq 0$. However, the problem posed is the converse: ‘show that *if* $a^2 - 4b^3 \geq 0$, *then* there exist real numbers x and y such that $a = xy(x + y)$ and $b = xy$ ’.

So this attempt has proved the converse, though the argument would have been slightly better presented if the student had explicitly stated ‘assume that x and y are real numbers with $a = xy(x + y)$ and $b = xy$ ’.

The correct option is therefore C.

Question 12

- I We can think of the integral $\int_0^x f(t) dt$ as being the (signed) area under the graph of $y = f(x)$ from 0 to x . Since $f(x) \geq g(x)$ for all $x \geq 0$, the area under the graph of $y = f(x)$ will also be greater than that under the graph of $y = g(x)$, so this statement is true.

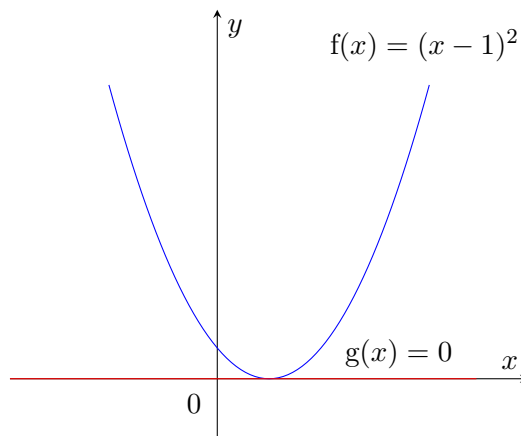
You might be concerned about values of x for which $g(x)$ is negative. We could therefore demonstrate this algebraically as well.

We have

$$\int_0^x f(t) dt - \int_0^x g(t) dt = \int_0^x (f(t) - g(t)) dt.$$

Since $f(t) - g(t) \geq 0$ for all $x \geq 0$, we are integrating a non-negative function, and so the integral must be non-negative, that is to say, we must have $\int_0^x f(t) dt \geq \int_0^x g(t) dt$.

- II It is not clear why $f(x) \geq g(x)$ should imply that the gradient (derivative) of $f(x)$ is greater than that of $g(x)$, and it is relatively straightforward to find a counterexample. Here is one:



In this example, $f(x) \geq g(x)$ for all x , but $f'(0) < 0$ while $g'(x) = 0$ for all x .

- III This is not necessarily true, as we could add a constant to $g(x)$ without changing $g'(x)$. For example, we could have $f'(x) = g'(x) = 1$ for all x , but $f(x) = x$ while $g(x) = x + 1$, so $f(x) < g(x)$ for all x .

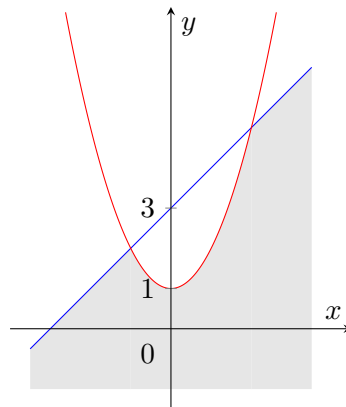
The correct option is therefore B.

Question 13

Manipulating inequalities algebraically requires care. For example, we cannot (in general) multiply two inequalities together and obtain a valid inequality (which is what statement II has done). Neither can we subtract two inequalities (which is where statement I comes from).

In this case, one approach is to sketch the region R described by the pair of inequalities and then determine whether the inequalities in statements I–III are satisfied in the whole of R . Another approach is to search for counterexamples for each statement, that is, values of x and y that do not satisfy the statement but do satisfy the original inequalities. This, however, might be harder to do, and if the inequalities are true, it may take a while to discover this. So let us sketch the given inequalities.

The inequality $y - x < 3$ can be rearranged as $y < x + 3$, so it is the region below $y = x + 3$. Similarly, $y - x^2 < 1$ can be rearranged as $y < x^2 + 1$, so it is the region below the quadratic $y = x^2 + 1$. The region R is therefore the shaded area shown here (excluding the boundaries):



We can now consider the three statements.

- I It is clear that this is not true in general, as the region R extends in all directions. For example, the point $(10, 0)$ lies in R (it is easy to check that this satisfies the original inequalities: $0 - 10 < 3$ and $0 - 10^2 < 1$) but $x = 10$ does not satisfy $-1 < x < 2$.
- II If we take $x = 0$, the inequality becomes $y^2 < 3$, but $(0, -2)$ is in R and $(-2)^2 = 4$. So this statement is not true for all points in R .
- III We could take, for example, $(6, 6)$ as a point in R with $y \geq 5$, so this statement is not true in general.

The correct option is therefore A.

Question 14

Writing $X = 2^x$ and $Y = \log_2 y$ gives the simultaneous equations

$$\begin{aligned} pX + Y &= 2 \\ X + Y &= 1 \end{aligned}$$

Subtracting gives $(p - 1)X = 1$, so as long as $p \neq 1$, we have $X = \frac{1}{p-1}$ and so $Y = \frac{p-2}{p-1}$.

Now the equation $\log_2 y = \frac{p-2}{p-1}$ gives a real value for y for any value of $p \neq 1$ (specifically $y = 2^{(p-2)/(p-1)}$), but the equation $2^x = \frac{1}{p-1}$ has a solution if and only if $\frac{1}{p-1} > 0$, which is if and only if $p > 1$.

Therefore the original simultaneous equations have a real solution (x, y) if and only if $p > 1$, which is option C.

Question 15

Approach 1

We start by completing the square to find the centre and radius of the circle:

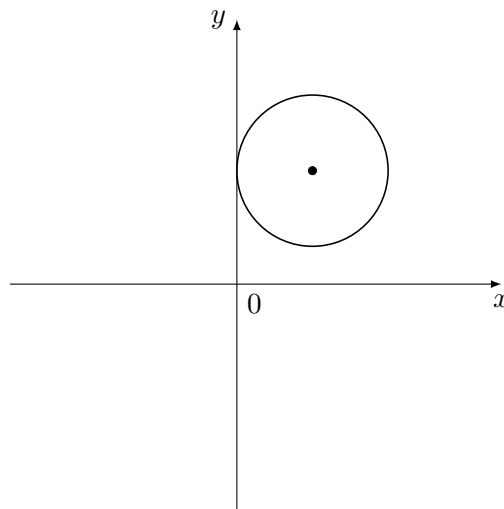
$$(x + \frac{1}{2}a)^2 - (\frac{1}{2}a)^2 + (y + \frac{1}{2}b)^2 - (\frac{1}{2}b)^2 + c = 0$$

which gives

$$(x + \frac{1}{2}a)^2 + (y + \frac{1}{2}b)^2 = (\frac{1}{2}a)^2 + (\frac{1}{2}b)^2 - c$$

so the circle has centre $(-\frac{1}{2}a, -\frac{1}{2}b)$ and radius $\sqrt{(\frac{1}{2}a)^2 + (\frac{1}{2}b)^2 - c}$.

Now let us sketch a circle tangent to the y -axis:



We can see that the circle is tangent to the y -axis if and only if its radius is equal to the distance of its centre from the y -axis. This is *not* the same as saying the radius is equal to the x -coordinate of the centre, as the centre may have a negative x -coordinate. As an equation, this condition is then

$$\sqrt{(\frac{1}{2}a)^2 + (\frac{1}{2}b)^2 - c} = |-\frac{1}{2}a|$$

which, on squaring, is equivalent to

$$(\frac{1}{2}a)^2 + (\frac{1}{2}b)^2 - c = (\frac{1}{2}a)^2$$

(This is equivalent to the previous equation because both sides of the first equation are non-negative.)

Subtracting $(\frac{1}{2}a)^2$ from both sides gives $\frac{1}{4}b^2 - c = 0$ or $b^2 = 4c$, so this condition is necessary and sufficient for the circle to be tangent to the y -axis, which is option B.

Approach 2

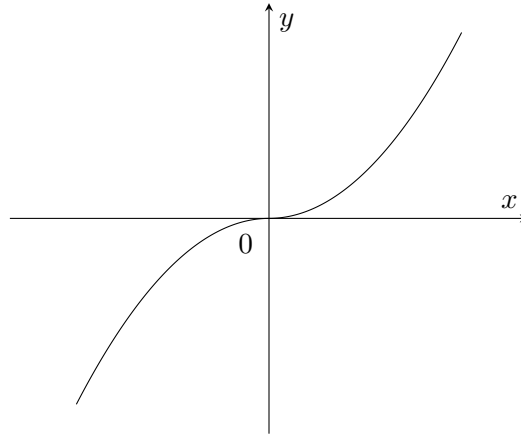
The circle is tangent to the y -axis, $x = 0$, if and only if the pair of simultaneous equations $x^2 + ax + y^2 + by + c = 0$ and $x = 0$ has a repeated root.

Substituting $x = 0$ into the circle equation gives $y^2 + by + c = 0$, which has discriminant $b^2 - 4c$. So there is a repeated root if and only if $b^2 - 4c = 0$, which is equivalent to option B.

Question 16

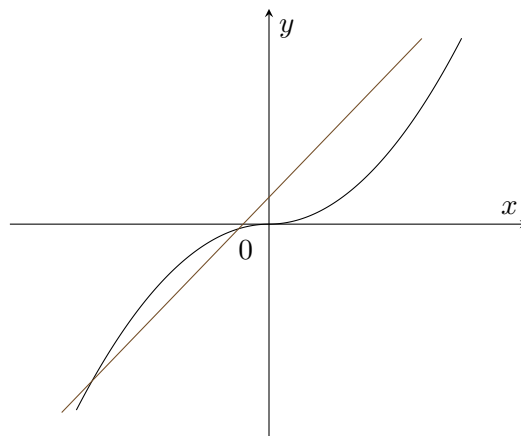
Approach 1: graph sketching

We are looking for the intersections of the graphs of $y = x|x|$ and $y = px + q$. The latter is just a straight line graph, while the former is $y = x^2$ when $x \geq 0$ and $y = -x^2$ when $x < 0$:



This graph looks quite similar to the graph of $y = x^3$: it has a point of inflexion at the origin. Any straight line will meet this graph at least once: if $q > 0$, then it will meet it for some $x > 0$, and if $q < 0$, then it will meet it for some $x < 0$. A straight line can meet the graph twice if it is tangent to the curve somewhere, and it can meet it three times, for example $y = x$.

But a straight line cannot meet the curve four times. It can meet the curve at most twice for $x \geq 0$, and at most twice for $x < 0$ (as each half of the curve is a quadratic, and a straight line and quadratic can meet at most twice). But if a straight line meets the quadratic twice in the region $x < 0$, then it must have $q > 0$, for example:



It can then meet the curve only once in the region $x > 0$. The same applies if it meets the curve twice in the region $x > 0$. And if the line passes through the origin, then it can meet the curve in at most two other locations.

So the correct answer is option E.

Approach 2: algebraic

If x is a solution, then either $x \leq 0$ and $-x^2 = px + q$ or $x > 0$ and $x^2 = px + q$; we can rewrite these as $x^2 + px + q = 0$ and $x^2 - px - q = 0$ respectively.

If $q < 0$, then the quadratic equations both have positive discriminant. The first quadratic has roots of opposite signs (as $\sqrt{p^2 - 4q} > p$), so there is exactly one negative solution. The second quadratic (corresponding to $x > 0$) may have 0, 1 or 2 positive solutions, giving 1, 2 or 3 solutions in total.

If $q > 0$, exactly the opposite is true, and we still end up with 1, 2 or 3 solutions.

If $q = 0$, then $x = 0$ is a root of both quadratics, and there are 0, 1 or 2 other roots.

Thus the possible values are 1, 2 and 3.

Examples with these numbers are: $p = q = 0$; $p = -1, q = 0$, and $p = -2, q = 1$ respectively.

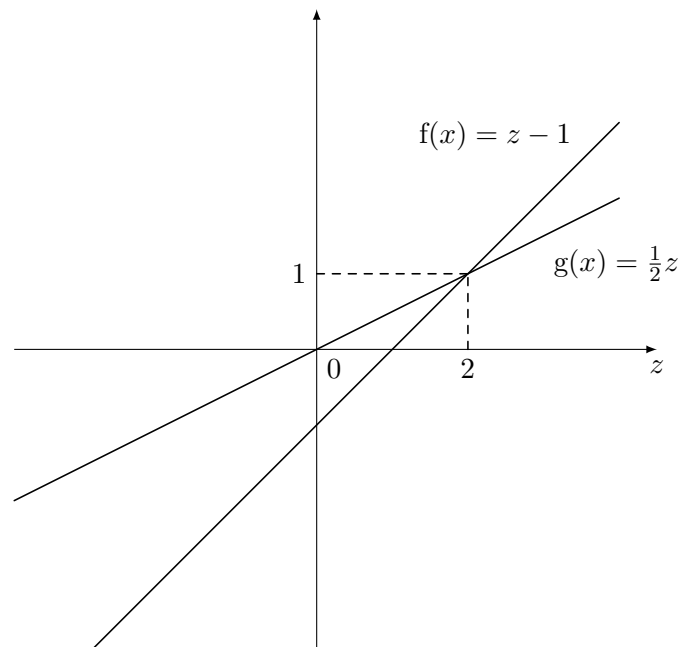
Question 17

The expressions for $f(x)$ and $g(x)$ look quite complicated, so we start by simplifying them using the laws of logarithms:

$$\begin{aligned} f(x) &= \log_2(\log_2 \sqrt{x}) \\ &= \log_2\left(\frac{1}{2} \log_2 x\right) \\ &= -1 + \log_2(\log_2 x) \\ g(x) &= \log_2(\sqrt{\log_2 x}) \\ &= \frac{1}{2} \log_2(\log_2 x) \end{aligned}$$

We now see that both $f(x)$ and $g(x)$ can be written as a function of $\log_2(\log_2 x)$. As $x > 1$, $\log_2 x > 0$ and $\log_2(\log_2 x)$ can take any real value.

Let us write $z = \log_2(\log_2 x)$, so that we have $f(x) = z - 1$ and $g(x) = \frac{1}{2}z$. We can sketch a graph showing these two functions of z :



We can calculate the intersection of $y = z - 1$ and $y = \frac{1}{2}z$, obtaining $z = 2$, $y = 1$.

We can then see from the graph that $f(x) \geq g(x)$ when both are at least 1, and $f(x) \leq g(x)$ when both are at most 1, so either $f(x) \geq g(x) \geq 1$ or $f(x) \leq g(x) \leq 1$ (or both), which is option F.

(Note that the 'or' in the options is inclusive 'or', so we are allowed to have both of the inequalities being true.)

Question 18

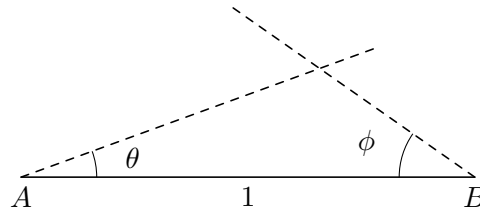
Since we are only given a condition on $\sin A$, the angle A could be acute or obtuse, and similarly for angle B .

Let us write $\theta = \sin^{-1} x$ and $\phi = \sin^{-1} y$ for the acute angles; the respective obtuse angles are then $180^\circ - \theta$ and $180^\circ - \phi$.

As we are given $x < y$, we have $\theta < \phi$.

We can now consider the four possible cases.

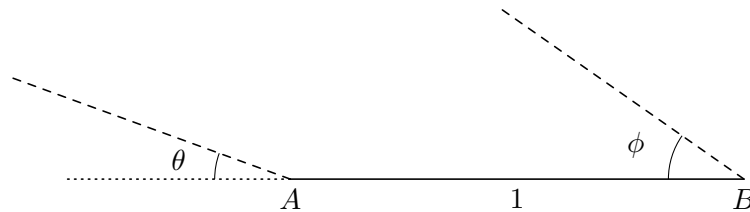
- Case 1: both A and B are acute. Then we have this situation:



We therefore know the three angles of the triangle ABC (as given two, we automatically know the third) and one side, and this determines the triangle uniquely (regarding congruent triangles as the same).

So we always obtain exactly one triangle from this case.

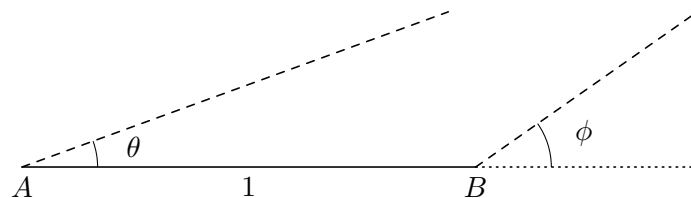
- Case 2: A is obtuse and B is acute. Then $A = 180^\circ - \theta$ and we have this situation:



Since $\theta < \phi$, the two lines (rays) from A and B shown in this diagram will never meet. Alternatively, we could note that $A + B = 180^\circ - \theta + \phi$, and since $\phi - \theta > 0$, this means that the angles A and B sum to more than 180° .

Therefore this case can never produce a triangle.

- Case 3: A is acute and B is obtuse. Then $B = 180^\circ - \phi$ and we have this situation:



Since $\theta < \phi$, the two lines (rays) from A and B shown will eventually meet, and so we get a unique triangle. Alternatively, we have $A + B = \theta + 180^\circ - \phi = 180^\circ - (\phi - \theta)$, and since $\phi - \theta > 0$, this means that the angles A and B sum to less than 180° , so they can be two angles of a triangle.

So we always obtain exactly one triangle from this case.

- Case 4: A and B are both obtuse. This is impossible in a triangle.
Therefore this case can never produce a triangle.

In summary, cases 1 and 3 always produce a unique triangle, and cases 2 and 4 never produce a triangle, so the student can always draw exactly two different triangles, regardless of the values of x and y .

Therefore the correct answer is C.

Question 19

We can rewrite the given equation as

$$\sin \theta \sqrt{(1 + \sin \theta)(1 - \sin \theta)} + \cos \theta \sqrt{(1 + \cos \theta)(1 - \cos \theta)} = 0$$

and then multiply out the brackets to give

$$\sin \theta \sqrt{1 - \sin^2 \theta} + \cos \theta \sqrt{1 - \cos^2 \theta} = 0$$

which simplifies to

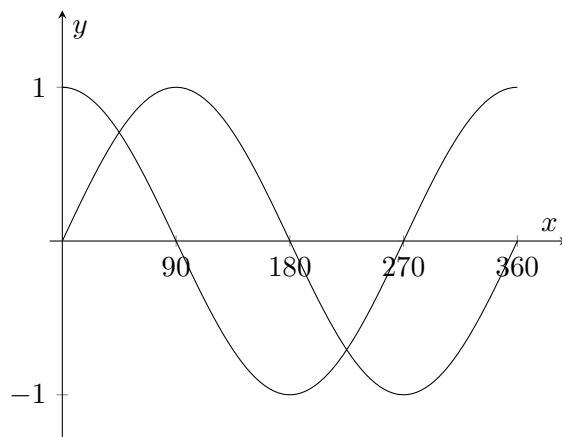
$$\sin \theta \sqrt{\cos^2 \theta} + \cos \theta \sqrt{\sin^2 \theta} = 0.$$

It would be easy at this point to make the mistake of simplifying $\sqrt{\cos^2 \theta}$ to $\cos \theta$, but this would only work in the case that $\cos \theta \geq 0$ as square roots are always positive. We can, though, write $|\cos \theta|$ for this square root: we always have $\sqrt{x^2} = |x|$. Therefore we can rewrite the equation as

$$\sin \theta |\cos \theta| + \cos \theta |\sin \theta| = 0.$$

If both $\sin \theta$ and $\cos \theta$ are positive, then this becomes $2 \sin \theta \cos \theta = 0$, which cannot be true. Likewise, if both $\sin \theta$ and $\cos \theta$ are negative, this becomes $-2 \sin \theta \cos \theta = 0$, which again cannot be true. However, if one is positive and one is negative, say $\sin \theta > 0$ and $\cos \theta < 0$, this becomes $-\sin \theta \cos \theta + \sin \theta \cos \theta = 0$, which is certainly true. The same holds if $\sin \theta < 0$ and $\cos \theta > 0$. Finally, if either $\sin \theta = 0$ or $\cos \theta = 0$, the equation is also satisfied.

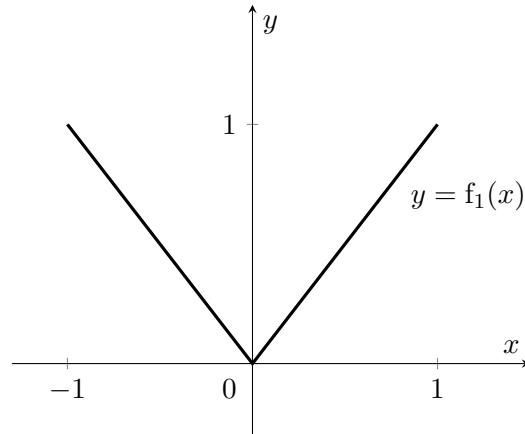
We therefore need to count the number of given angles for which $\sin \theta$ and $\cos \theta$ have opposite signs or one of them is zero. A quick sketch helps here:



We see that the relevant ranges are $90^\circ \leq \theta \leq 180^\circ$ and $270^\circ \leq \theta \leq 360^\circ$; $x = 0^\circ$ would also work, but is not one of the angles in the given list. There are 91 angles in each of the two ranges (remembering to include both endpoints), so there are 182 angles in total, which is option F.

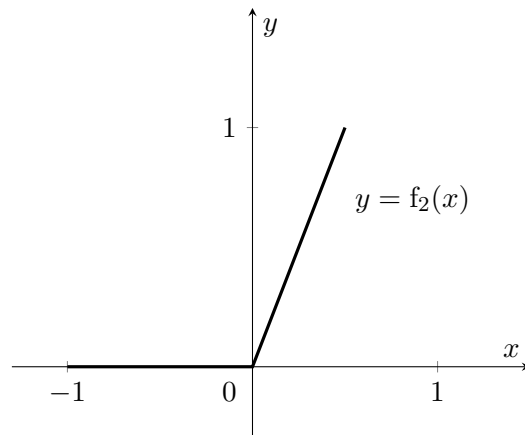
Question 20

It is helpful to get a feeling for what this sequence of functions looks like, so we will sketch the first few. We are only interested in the range from -1 to 1 (because that is what the integral asks for). The first function, $f_1(x) = |x|$ is standard:

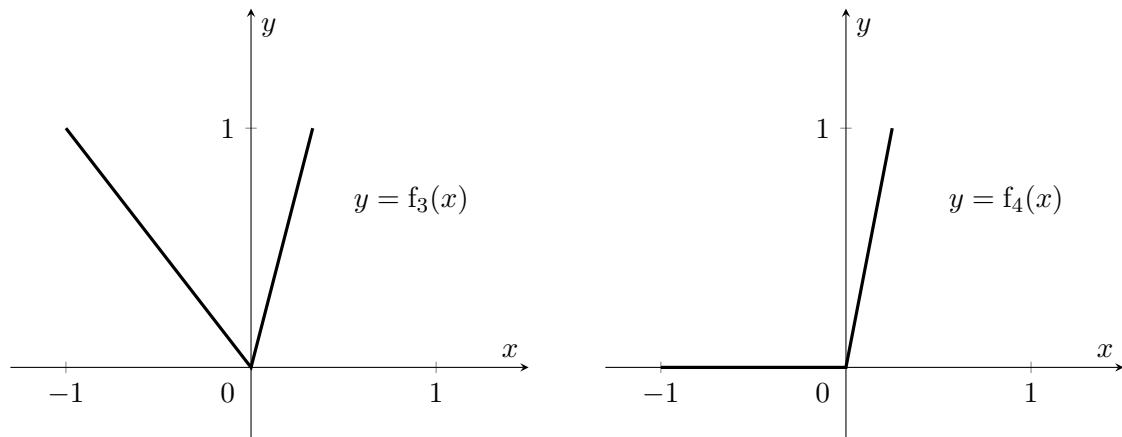


Note that when $x < 0$, this is the graph of $y = -x$, and when $x \geq 0$, this is the graph of $y = x$.

We next have $f_2(x) = |f_1(x) + x|$. We can calculate this on the two regions $x < 0$ and $x \geq 0$ separately: when $x < 0$, we get $f_2(x) = |-x + x| = 0$, and when $x > 0$, we get $f_2(x) = |x + x| = 2x$, giving this graph:



We can determine $f_3(x)$ and $f_4(x)$ in a similar way:



Here, the graphs for $x \geq 0$ are $y = 3x$ and $y = 4x$ respectively.

We can see a pattern forming: when n is odd, the graph of $y = f_n(x)$ looks like $y = f_3(x)$, with $y = -x$ for $x < 0$ and $y = nx$ for $x \geq 0$, and when n is even, it looks like $y = f_4(x)$, with $y = 0$ for $x < 0$ and $y = nx$ for $x \geq 0$.

We are interested in $f_{99}(x)$, which looks like $y = f_3(x)$: it is $y = -x$ for $x < 0$ and $y = 99x$ for $x \geq 0$. We can calculate the integral $\int_{-1}^1 f_{99}(x) dx$ by finding the area under the graph, which is just the sum of two triangles.

The area between $x = -1$ and $x = 0$ is $\frac{1}{2} \times 1 \times 1$ and the area between $x = 0$ and $x = 1$ is $\frac{1}{2} \times 1 \times 99$, so the integral equals $\frac{1}{2} \times 1 + \frac{1}{2} \times 99 = \frac{1}{2} \times 100 = 50$; the correct option is therefore E.

This document was initially designed for print and as such does not reach accessibility standard WCAG 2.1 in a number of ways including missing text alternatives and missing document structure.

If you need this document in a different format please email admissionstesting@cambridgeassessment.org.uk telling us your name, email address and requirements and we will respond within 15 working days.

This document contains maths equations and symbols which at present we are not able to produce adequately for accessibility. We are working to improve this.

We are Cambridge Assessment Admissions Testing, part of the University of Cambridge. Our research-based tests provide a fair measure of skills and aptitude to help you make informed decisions. As a trusted partner, we work closely with universities, governments and employers to enhance their selection processes.

Cambridge Assessment
Admissions Testing
The Triangle Building
Shaftesbury Road
Cambridge
CB2 8EA
United Kingdom

Admissions tests support: admissionstesting.org/help